TEXORA · LECTURE SHEET

Polymer Degradation and Environmental Aspects

How polymer chains are attacked and broken — by heat, oxygen, light, water, chemicals and microbes — what that does to fibres in service, and the environmental bill: waste, microfibres, recycling and biodegradable alternatives.

TE-2107 · POLYMER SCIENCE AND ENGINEERING · TOPIC 6 · LEVEL 2 TERM 1

1. What degradation is

Degradation is any change in a polymer's structure that worsens its properties: chains are cut (**chain scission**), so molecular weight, strength and elongation fall; or chains are joined (**cross-linking**), so the material hardens and embrittles; or side groups are stripped, changing colour and chemistry. Some polymers **depolymerise ("unzip")** — PMMA and PTFE revert largely to monomer when overheated — while most suffer random scission, each break converting one long chain into two shorter ones. Because strength depends on chain length (Topics 1 and 4), even a few breaks per chain are serious: halving the average chain length can cost far more than half the toughness.

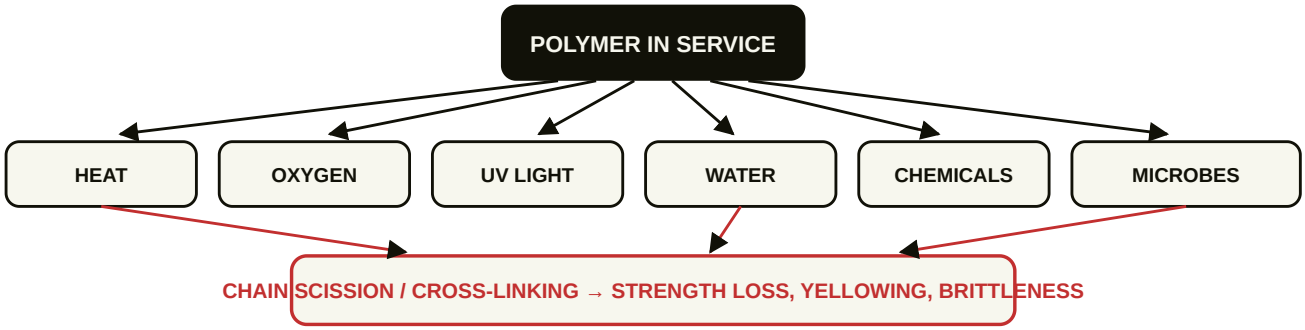


Figure 1 — The six attacking agents of degradation and their common result: a fall in molecular weight and with it in strength and service life. Stabilisers (Topic 5) are chosen to block the specific agent a product will meet.

2. The main degradation routes

2.1 Thermal degradation

Heat alone, without oxygen, breaks the weakest bonds first. PVC unzips HCl from about 180 °C upward; PET and nylon survive their melt-processing window (280–290 °C) only briefly and only

when dry — residence time in the extruder is a controlled quality variable. Each chain scission in the melt shows up later as lower intrinsic viscosity and brittle filament.

2.2 Oxidative degradation (thermo-oxidation)

Oxygen plus heat (or light) runs a radical chain reaction: a radical forms on a chain, adds O_2 to make a peroxy radical, which steals hydrogen from the next chain, creating a hydroperoxide and a new radical. Hydroperoxides split into more radicals, so the attack accelerates (auto-oxidation). Consequences: yellowing, surface cracking, loss of strength — the brittle PP rope left in the sun is the classic case. Antioxidants exist to break this cycle (Topic 5).

2.3 Photo-degradation

Ultraviolet photons (290–400 nm) carry enough energy to break C–C and C–H bonds, especially at carbonyl impurities and catalyst residues that absorb UV. The surface layers degrade first: nylon loses roughly half its strength after prolonged strong sunlight; polyester is better but not immune; cotton and viscose also tender in sunlight. Photo-oxidation combines routes 2.2 and 2.3 and is the main life-limiter of outdoor textiles — awnings, ropes, fishing nets, geotextiles.

2.4 Hydrolytic degradation

Water attacks the very bonds that condensation polymerisation made: the ester links of PET and the amide links of nylon are cleaved by H^+ or OH^-, faster when hot. Practical consequences: PET chips must be dried to very low moisture before melt spinning, or the melt hydrolyses and IV collapses; nylon is attacked by strong acids; hot strong alkali surface-hydrolyses PET — deliberately used to soften polyester fabric ("weight reduction" finishing) and accidentally destructive in careless washing. Addition polymers with all-carbon backbones (PE, PP, PS) are immune to hydrolysis.

2.5 Chemical, mechanical and ozone attack

Strong acids dissolve nylon and tender cotton (acid hydrolysis of cellulose); strong oxidants (bleach) attack most fibres if overdosed; ozone cracks strained rubber by attacking its double bonds. **Mechanical degradation** is real too: violent milling, high-shear processing and even long ultrasonic exposure tear chains apart, and every recycle pass through an extruder shortens chains a little further.

2.6 Biodegradation

Microbes and their enzymes digest natural polymers readily — cotton, wool and viscose biodegrade — and can attack some synthetics extremely slowly. The same durability that makes PE or PET useful makes them persist in the environment for decades to centuries. Engineered biodegradable polymers (PLA, PHA, PBAT) carry linkages microbes can cleave, but most still need industrial composting conditions (heat, moisture, microbial load) to degrade in reasonable time.

FIBRE POLYMER	CHIEF WEAKNESSES	RELATIVE OUTDOOR LIFE
Cotton / viscose (cellulose)	Acid, microbes, sunlight tendering	Short–moderate; rots when damp
Nylon	UV, strong acid, oxidation	Moderate; UV is the limiter
PET polyester	Hot alkali, hydrolysis in the melt	Good; among the better UV performers of the synthetics
Acrylic	Few; best UV resistance of common fibres	Long — hence awnings and outdoor upholstery
Polypropylene	Oxidation and UV (unstabilised, very poor)	Short unless stabilised; then moderate

3. Environmental aspects

3.1 The waste problem

Global fibre production is of the order of a hundred million tonnes a year, and the large majority of textile products eventually reach landfill or incineration, with only a small fraction — roughly a tenth or less — collected for recycling. Synthetic fibres do not rot in landfill; natural fibres do, but anaerobically, releasing methane. Incineration recovers energy but returns fossil carbon (for synthetics) to the air as CO_2.

3.2 Microfibres and microplastics

Every wash of synthetic clothing sheds tiny fibre fragments — **microfibres** (a subset of microplastics) — which pass through sewage treatment in large numbers and accumulate in rivers and oceans, entering food chains. Staple-fibre fabrics shed more than continuous-filament ones; tighter constructions and pre-washing at the factory reduce release. This is now a design parameter of synthetic textiles, not an afterthought, and legislation on washing-machine filters and fibre-shedding limits is emerging in several markets.

3.3 Recycling routes

- **Mechanical recycling** — shredding, melting and re-spinning. Works well for clean single-polymer streams (recycled PET from bottles into fibre is the big success), but each pass shortens chains (Topic 4: IV falls) and blends, dyes and finishes contaminate the stream; textile-to-textile mechanical recycling is limited by fibre shortening.
- **Chemical recycling** — depolymerising back to monomer or oligomer and re-polymerising to virgin-quality polymer: PET by glycolysis, methanolysis or hydrolysis; nylon 6 back to caprolactam;

cotton-rich waste can be dissolved and regenerated as new cellulosic fibre. Energy-costly but quality-restoring, and the only route that fully escapes degradation history.

- **Design for recycling** — mono-material garments, avoiding inseparable blends and problematic finishes, are the precondition for either route at scale.

3.4 Biodegradable and bio-based polymers

PLA (polylactic acid) is polymerised from lactic acid obtained by fermenting plant sugars; it is bio-based *and* industrially compostable, melt-spinnable, with a melting point around 170 °C — usable in packaging, nonwovens and some apparel, though brittle and heat-sensitive. Bio-based is not the same as biodegradable: bio-PET made from plant-derived glycol is chemically ordinary PET and persists like it. Honest labelling distinguishes the two claims.

Glossary

Chain scission — cutting of a polymer backbone bond, reducing molecular weight. **Depolymerisation (unzipping)** — successive loss of monomer units from a chain end. **Auto-oxidation** — the self-accelerating radical oxidation cycle via hydroperoxides. **Hydrolysis** — bond cleavage by water, cutting ester and amide links. **Microfibres** — sub-5 mm fibre fragments shed from textiles, a class of microplastics. **Glycolysis** — chemical recycling of PET using ethylene glycol to produce BHET oligomers. **PLA** — polylactic acid, a compostable polyester from fermented plant sugars.

Key points

• Degradation = chain scission, cross-linking or group loss; strength tracks molecular weight, so small chemical damage has large mechanical effects.
• Heat, oxygen and UV act together outdoors; hydrolysis attacks the condensation polymers (PET, nylon) exactly where they were joined.
• Condensation polymers must be processed dry; carbon-backbone polymers cannot hydrolyse but oxidise instead (PP worst).
• Synthetics persist in landfill and shed microfibres in washing — durability is an environmental cost as well as a benefit.
• Recycling escapes: mechanical (quality falls each cycle) or chemical (back to monomer, virgin quality); biodegradable ≠ bio-based.

Practice

1. Wet PET chips are melt-spun without proper drying and the filament keeps breaking. Explain the chemistry of the failure and name the property measurement that would have warned the plant.

2. An unstabilised PP rope and a stabilised one are used outdoors side by side. After a year the first snaps with little load while the second is sound. Describe the reaction cycle that destroyed the first rope and the two points where the stabiliser package interrupts it.
3. Why can nylon 6 be chemically recycled to virgin quality more easily than polyethylene? Answer in terms of bond chemistry.
4. A garment is labelled "made from bio-based polyester". A customer assumes it will biodegrade like cotton. Correct the misunderstanding in two sentences.
5. Give three design or production measures that reduce microfibre shedding from a fleece jacket.

Answer key

1. At 280 °C the water in the chips hydrolyses ester bonds along the PET chains, cutting molecular weight (IV falls sharply). Shorter chains give a weak, brittle melt and filament. The warning measurement is intrinsic viscosity [η] of the melt/chips (or the melt's end-group count) — a falling IV before spinning signals hydrolysis.
2. Photo-oxidation: UV creates radicals on the PP chain (tertiary carbons are most vulnerable); radicals add O_2 → peroxy radicals → hydroperoxides, which split to spawn more radicals — an accelerating cycle of chain scission. Stabilisers interrupt it (i) at initiation, by absorbing UV or screening it, and (ii) during propagation, by antioxidants/HALS donating hydrogen or trapping radicals.
3. Nylon 6's amide links can be cleaved reversibly — heating with steam or catalyst unzips the chain back to caprolactam monomer, which is purified and repolymerised. Polyethylene's backbone is pure C–C bonds with no hydrolysable or reversible link; it can only be cracked thermally into mixed hydrocarbons, not cleanly back to ethylene.
4. "Bio-based" describes where the carbon came from (plants rather than oil), not what happens after disposal. Bio-PET is chemically identical to ordinary PET, so it persists in the environment like any polyester; only polymers designed with microbe-cleavable linkages and composting conditions, such as PLA, actually biodegrade.
5. Any three: use continuous-filament yarn instead of staple for the face fabric; tighter/higher-density construction; singeing or shearing and thorough pre-washing at the factory to remove loose fibres before sale; resin or surface finishes that bind fibres; (consumer-side) wash less often, gentler cycles, filter bags/filters.
